



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO T.A.M.S.K. FUEL SERVICES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

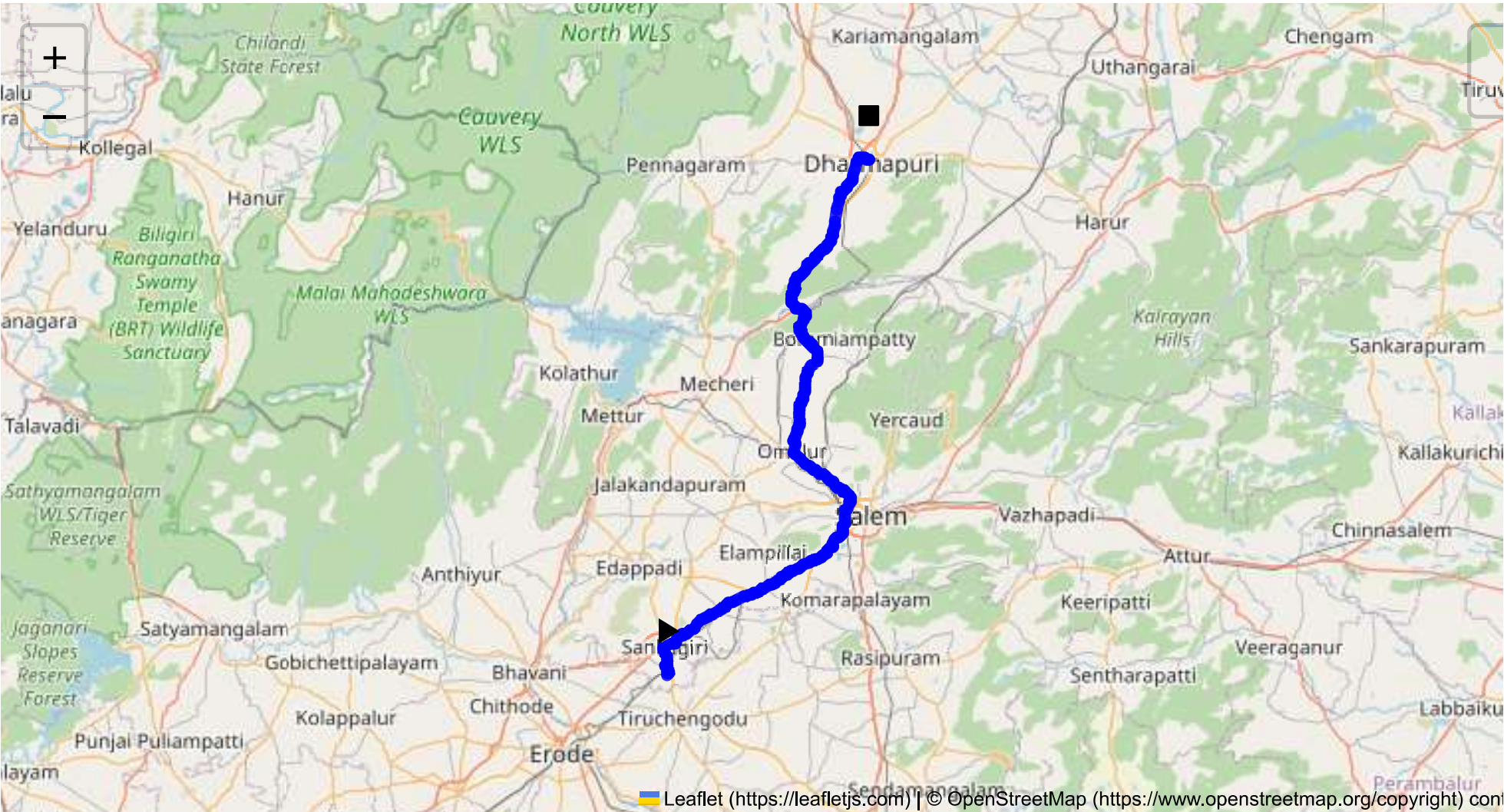
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 107.04 km
Estimated Duration: 2.1 hours
Adjusted Duration (Heavy Vehicle): 2.6 hours
Start: (11.4381, 77.8734)
End: (12.139506, 78.151982)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route between Sangagiri and Annai Sathya Nagar spans approximately 107.04 kilometers in Tamil Nadu, traversing through a mix of rural and semi-urban areas. It typically takes about 2.05 hours for heavy vehicles. The route likely involves major highways and rural roads, necessitating awareness of varying road conditions.

2. Typical Weather Conditions

Tamil Nadu experiences tropical weather. Summers (March to June) can be very hot, while the monsoon season (June to September) might bring heavy or moderate rainfall, potentially leading to slippery roads and reduced visibility. The region might also be prone to cyclones and subsequent heavy downpours in late monsoon or post-monsoon seasons (October to December).

3. Traffic Patterns

Traffic congestion is more prevalent during peak hours, typically morning (7:30 AM - 10:00 AM) and evening (5:00 PM - 8:00 PM). Key areas of congestion are likely on highways entering or exiting larger towns, particularly near schools, markets, or industrial zones. Be cautious around urban centers where traffic build-ups are common.

4. Assessment of Road Quality and Infrastructure

The primary highways may be well-maintained, but transitioning to rural roads can result in variable road conditions with occasional potholes or narrow stretches. Look out for ongoing construction projects that could temporarily affect road quality and create detours or delays.

5. Suggestions for Alternative Routes

For emergencies, consider state highways parallel to your main route. Depending on your precise trajectory, NH44 or SH20 might serve as alternatives. However, verify local data for any updates on road closures or construction work that could affect secondary route viability.

6. Local Regulations on Hazardous Material Transport

Tamil Nadu has stringent regulations regarding the transport of hazardous materials, focusing on driver credentials, specific route permissions, and travel times (daylight hours for higher safety). Ensure compliance with signage and load restrictions, particularly in and around populated areas.

7. Historical Incidents

Past incidents involving heavy vehicles in Tamil Nadu often relate to poor weather conditions, road infrastructure failures, or human error. Although specific instances are not detailed, maintain caution, particularly during adverse weather and at night.

8. Environmental Considerations

Be aware of sensitive areas, such as water bodies and nature reserves, where spillage of hazardous materials could have significant impacts. Environmental zones might have restrictions or detours that need to be observed.

9. Communication Coverage

Mobile network coverage is generally reliable along major routes but may be inconsistent in rural or remote stretches. Plan communications accordingly, and consider alternative methods (e.g., satellite phones) where coverage is weak.

10. Estimated Emergency Response Times

Emergency response in urban areas of Tamil Nadu is relatively quick (30-45 minutes), but rural segments may experience longer times (up to 1.5 hours) due to remote locations and access challenges. Familiarize with local emergency numbers and contacts before departure.

12. Overall Summary of Risk Assessment

This journey involves various risks typical to regional road transport, such as weather impacts, infrastructure quality, and traffic congestion. By preparing for these variables—preferably by traveling during off-peak hours and in stable weather conditions—the risks can be significantly mitigated. Continuous adherence to local regulations and maintaining open communication with emergency services will further enhance safety. Regular updates with local traffic advisories can also prove vital in optimizing route management, ultimately ensuring a safer journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	Medium	11.44900, 77.87406	30 KM/Hr
3	Turn	Medium	11.45000, 77.87213	30 KM/Hr
4	Turn	High	11.45028, 77.87202	15 KM/Hr
5	Turn	Medium	11.45309, 77.87325	30 KM/Hr
6	Turn	Medium	11.46400, 77.87323	30 KM/Hr
7	Turn	Medium	11.46765, 77.87037	30 KM/Hr
8	Turn	High	11.47403, 77.86841	15 KM/Hr
9	Turn	High	11.49149, 77.89601	15 KM/Hr
10	Turn	High	11.49224, 77.89601	15 KM/Hr
11	Turn	Medium	11.64180, 78.11808	30 KM/Hr
12	Turn	Medium	11.64325, 78.11993	30 KM/Hr
13	Turn	Medium	11.69519, 78.10775	30 KM/Hr
14	Turn	Medium	11.75835, 78.05016	30 KM/Hr
15	Turn	Medium	11.87942, 78.08197	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
16	Turn	Medium	11.95818, 78.04978	30 KM/Hr
17	Turn	Medium	11.95876, 78.04990	30 KM/Hr
18	Turn	Medium	11.96035, 78.04946	30 KM/Hr
19	Turn	Medium	11.96304, 78.05136	30 KM/Hr
20	Turn	Medium	11.96399, 78.05135	30 KM/Hr
21	Turn	Medium	11.96458, 78.05084	30 KM/Hr
22	Turn	Medium	11.96496, 78.04988	30 KM/Hr
23	Turn	High	11.96580, 78.04923	15 KM/Hr
24	Turn	Medium	11.96675, 78.04939	30 KM/Hr
25	Turn	Medium	11.96791, 78.05070	30 KM/Hr
26	Turn	High	12.14053, 78.13964	15 KM/Hr
27	Turn	High	12.13966, 78.15289	15 KM/Hr
28	Blind Spot	Blind Spot	12.13948, 78.15300	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
7	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
8	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
10	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
13	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
16	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
20	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
21	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
22	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
23	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
25	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
26	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
27	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
28	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
29	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
32	pharmacy	MJ Medicals	11.7232728, 78.0698965	30 km/h	Medium
33	pharmacy	Salem Medicals	11.7343192, 78.0566844	30 km/h	Medium
35	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
36	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
37	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
38	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
39	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
40	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
42	hospital	Vijaya Hospital	12.1399758, 78.1532115	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
14	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
15	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
17	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
18	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
19	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
24	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
30	college	Govt Engineering College	11.7128537, 78.0877084	30 km/h	Medium
31	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
34	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
41	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45000, 77.87213



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45309, 77.87325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64180, 78.11808



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64325, 78.11993



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69519, 78.10775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75835, 78.05016



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95876, 78.04990



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96035, 78.04946



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96458, 78.05084



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96496, 78.04988



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.14053, 78.13964



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.13966, 78.15289



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.13948, 78.15300

